CSE 318 — Online 2

Section C1

Your local search uses **best-improvement**: at each step it looks at *every* vertex, computes the gain $\delta(v)$ of flipping it, and flips the vertex with the largest positive δ. It stops when no vertex has a positive δ.

Your task: Implement a **first-improvement** local search. Instead of scanning all vertices and picking the best, flip the **first** improving vertex you find:

1. At the start of each local search call, generate a **random permutation** of the vertices. This is the order in which you will scan them.
2. Go through the vertices in that order. For each vertex v, compute $\delta(v)$.
3. As soon as you find a vertex v with $\delta(v) > 0$, flip it immediately.
4. After a flip, **do not restart** from the beginning of the permutation. Continue scanning from where you left off, going around circularly.
5. Keep a counter of how many consecutive vertices you have checked without finding an improvement. When this counter reaches n (total number of vertices), stop — you are at a local optimum.

What to run and report:

- Run GRASP with first-improvement local search on at least 6 benchmark graphs.
- Run GRASP with your original best-improvement local search on the same graphs.
- Use the same number of GRASP iterations for both.
- Report: best cut found, and total time taken.
- Make a comparison table.